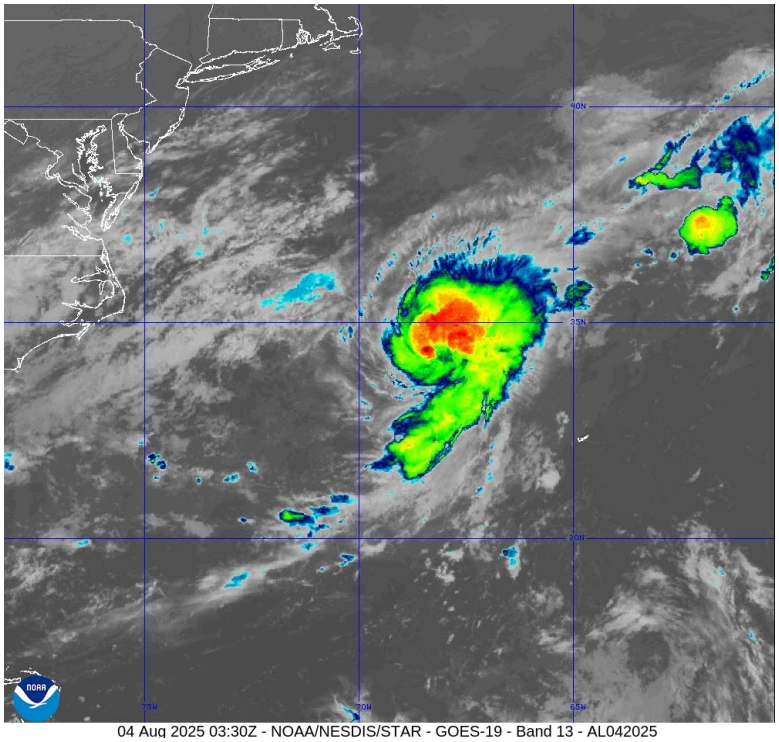
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

TROPICAL STORM DEXTER

(AL042025)

3–6 August 2025

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National Hurricane Center
23 October 2025



GOES-EAST INFRARED SATELLITE IMAGE OF TROPICAL STORM DEXTER AT 0330 UTC 4 AUGUST 2025. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Dexter was a tropical storm over the western Atlantic that did not affect land.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Dexter.

Tropical Storm Dexter

3-6 AUGUST 2025

BEST TRACK

The "best track[2]" positions and intensities for Tropical Storm Dexter are listed in Table 1. The best track chart of Dexter's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively.

There were no ship reports of winds of tropical storm force associated with Dexter. However, two NOAA buoys both recorded peak sustained winds of 31 kt near the system on 3 August and were helpful in the analysis of the best track.

Origin

Dexter originated from a stationary front over the western Atlantic near the southeast coast of the United States. A non-tropical low formed along the front around 1800 UTC 2 August offshore of the Carolinas, and ASCAT data that evening confirmed that the low-level wind circulation had become well defined. The portion of the front that extended to the west-southwest of the low dissipated during the morning of 3 August as gale-force winds developed in the southeast quadrant. Early that afternoon, convection quickly became better organized while the front that extended to the east-northeast of the low weakened. It is estimated that the cyclone shed this front and became Tropical Storm Dexter around 1800 UTC 3 August about 325 n mi west-northwest of Bermuda.

Peak Intensity and Minimum Pressure

Dexter's 50-kt peak intensity while it was a tropical cyclone occurred from 1200–1800 UTC 6 August and is supported by ADT, AiDT, DPRINT and SATCON intensity estimates from the University of Wisconsin-CIMSS, all of which peaked in the 50–55-kt range. At the same time, subjective Dvorak estimates from TAFB and SAB peaked at T 3.0/45 kt.

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[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison (CIMSS). Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Dexter.

After becoming extratropical at 0000 UTC 7 August, Dexter briefly strengthened into a hurricane-force extratropical cyclone early on 8 August when it was located about 410 n mi southeast of St. John's, Newfoundland.

The lowest minimum central pressure attained by Dexter while it was a tropical cyclone is estimated to be 999 mb, based on the Knaff-Zehr-Courtney (KZC) pressure-wind relationship.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Dexter.

FORECAST AND WARNING VERIFICATION

Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. The genesis of Dexter was not well anticipated. NHC first indicated a low (<40%) chance of genesis in the TWO 30 h prior to formation and raised the formation chances to medium 12 h prior to development. The majority of the global models depicted a weak, elongated low along a frontal boundary during their runs between 29 July and 1 August. On 2 August, the GFS started to consistently depict a more compact low in the surface fields with less frontal interaction, while the ECMWF did not show this until its 0000 UTC 3 August run, after the frontal low had already formed. Figure 4 shows composites of 7-day TWO genesis areas for each category prior to the formation of Dexter. All of the TWO ellipses correctly encompassed the location where genesis occurred.

A verification of NHC official track forecasts for Dexter is given in Table 3a. Official track forecast errors were lower than the mean official errors for the previous 5-yr period at all time periods. The NHC track errors were exceptionally low from 12–48 h, when the average track error for Dexter was less than half of the 5-yr means. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. The official forecast generally outperformed all available model guidance. The HMNI and CTCI were strong performers.

A verification of NHC official intensity forecasts for Dexter is given in Table 4a. Official intensity forecast errors were comparable to the mean official errors for the previous 5-yr period for the 12–36 h forecasts, and outperformed the 5-yr means at 48–60 h. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. Much of the guidance performed well, with HMNI and HFAI performing the best.

There were no coastal watches or warnings issued for Dexter.

ACKNOWLEDGEMENTS

Dr. Philippe Papin of NHC provided the genesis figure (Fig. 4).

Table 1. Best track for Tropical Storm Dexter, 3–6 August 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
02 / 1800	33.2	75.6	1013	30	extratropical
03 / 0000	32.9	74.6	1011	35	"
03 / 0600	33.1	73.5	1010	35	"
03 / 1200	33.4	72.2	1008	35	"
03 / 1800	33.7	71.0	1007	35	tropical storm
04 / 0000	34.0	69.9	1003	40	"
04 / 0600	34.4	68.8	1003	40	"
04 / 1200	35.0	67.9	1004	40	"
04 / 1800	35.7	66.9	1005	40	"
05 / 0000	36.4	65.9	1005	40	"
05 / 0600	37.0	64.9	1008	35	"
05 / 1200	37.7	63.9	1009	35	"
05 / 1800	38.2	62.8	1009	35	"
06 / 0000	38.8	61.8	1006	40	"
06 / 0600	39.3	60.5	1003	45	"
06 / 1200	39.6	59.0	999	50	"
06 / 1800	39.8	57.2	999	50	"
07 / 0000	39.9	55.2	999	50	extratropical
07 / 0600	40.4	53.2	998	55	"
07 / 1200	40.9	51.3	997	60	"
07 / 1800	41.4	48.9	996	60	"
08 / 0000	42.3	46.5	994	65	"
08 / 0600	43.6	45.0	989	70	"
08 / 1200	44.7	43.5	988	70	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
08 / 1800	45.2	41.6	991	60	"
09 / 0000	45.4	39.2	994	50	"
09 / 0600	45.7	36.7	997	45	"
09 / 1200	45.9	34.1	1000	40	"
09 / 1800	46.4	31.5	1002	35	"
10 / 0000	46.9	29.4	1004	30	"
10 / 0600	47.3	27.5	1005	30	"
10 / 1200	47.6	25.9	1007	30	"
10 / 1800	47.9	24.5	1008	30	"
11 / 0000	48.1	23.1	1008	30	"
11 / 0600	48.3	21.8	1008	30	"
11 / 1200	48.5	20.5	1009	25	"
11 / 1800	48.6	19.3	1009	25	"
12 / 0000	48.7	18.1	1009	25	"
12 / 0600	48.6	17.0	1008	25	"
12 / 1200	48.5	16.0	1008	25	"
12 / 1800	48.7	14.8	1008	25	"
13 / 0000	49.0	13.6	1008	25	"
13 / 0600	49.4	12.2	1008	25	"
13 / 1200					dissipated
06 / 1200	39.6	59.0	999	50	minimum pressure and maximum winds as a tropical cyclone

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the "Low" category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	30	30
Medium (40%-60%)	12	12
High (>60%)	0	0

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Tropical Storm Dexter, 3–6 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**14.0**	**17.1**	**24.3**	**26.7**	**56.0**			
OCD5	29.9	59.8	82.8	69.3	67.6			
Forecasts	10	8	6	4	2			
OFCL (2020-24)	23.0	34.3	45.8	58.7	73.5	89.8	128.7	185.4
OCD5 (2020-24)	45.1	95.7	150.9	203.1	252.7	295.4	366.2	426.6

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Tropical Storm Dexter, 3–6 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	14.1	14.8	20.8	13.3				
OCD5	26.4	54.6	74.1	28.2				
GFSI	29.8	75.8	132.6	192.7				
HWFI	29.3	56.4	87.0	79.3				
HMNI	**10.7**	15.8	24.7	36.7				
HFAI	26.4	57.4	99.2	111.9				
HFBI	28.0	64.0	104.3	157.3				
GDMI	24.2	59.9	107.8	140.4				
EMXI	14.8	23.1	33.9	75.4				
NVGI	22.6	65.7	111.6	200.6				
CTCI	**13.5**	19.0	22.7	**10.2**				
TVCA	15.1	30.1	52.0	74.5				
TVCX	14.8	28.2	49.0	69.1				
GFEX	18.4	42.1	76.8	132.3				
TVDG	15.3	30.1	53.0	74.5				
HCCA	**13.0**	21.2	31.8	72.5				
FSSE	**11.7**	21.9	30.6	70.4				
AEMI	37.4	87.3	155.5	234.1				
TABS	51.7	130.2	238.9	366.8				
TABM	14.5	34.8	53.0	65.4				
TABD	49.6	124.5	175.9	200.1				
Forecasts	8	6	4	2				

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Tropical Storm Dexter, 3–6 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	6.0	7.5	**8.3**	**5.0**	**5.0**			
OCD5	5.2	9.4	9.5	7.5	9.0			
Forecasts	10	8	6	4	2			
OFCL (2020-24)	5.1	7.3	8.6	10.0	10.5	10.9	12.4	13.6
OCD5 (2020-24)	6.8	10.6	13.8	16.5	17.9	19.2	21.4	19.9

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Tropical Storm Dexter, 3–6 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	6.2	7.5	7.5	7.5				
OCD5	**5.2**	8.7	**5.2**	**1.5**				
HWFI	**5.8**	**6.2**	9.5	10.0				
HMNI	**4.2**	**4.0**	**3.0**	**4.5**				
HFAI	**2.5**	**4.2**	**3.2**	**4.5**				
HFBI	**4.1**	**5.8**	**6.0**	9.0				
GDMI	**5.2**	**7.2**	**5.8**	8.0				
DSHP	**5.4**	**6.8**	**7.2**	13.5				
LGEM	6.6	10.3	13.8	18.0				
ICON	**4.9**	**6.2**	7.8	11.5				
IVCN	**4.2**	**5.5**	**5.0**	7.5				
IVDR	**4.2**	**5.5**	**4.5**	**6.5**				
CTCI	**5.4**	8.0	**4.2**	**3.5**				
GFSI	7.0	9.8	12.8	17.5				
EMXI	**5.9**	8.0	**4.0**	**5.0**				
HCCA	**4.5**	**5.3**	**3.5**	**7.0**				
FSSE	**4.1**	**5.2**	**5.5**	11.0				
Forecasts	8	6	4	2				

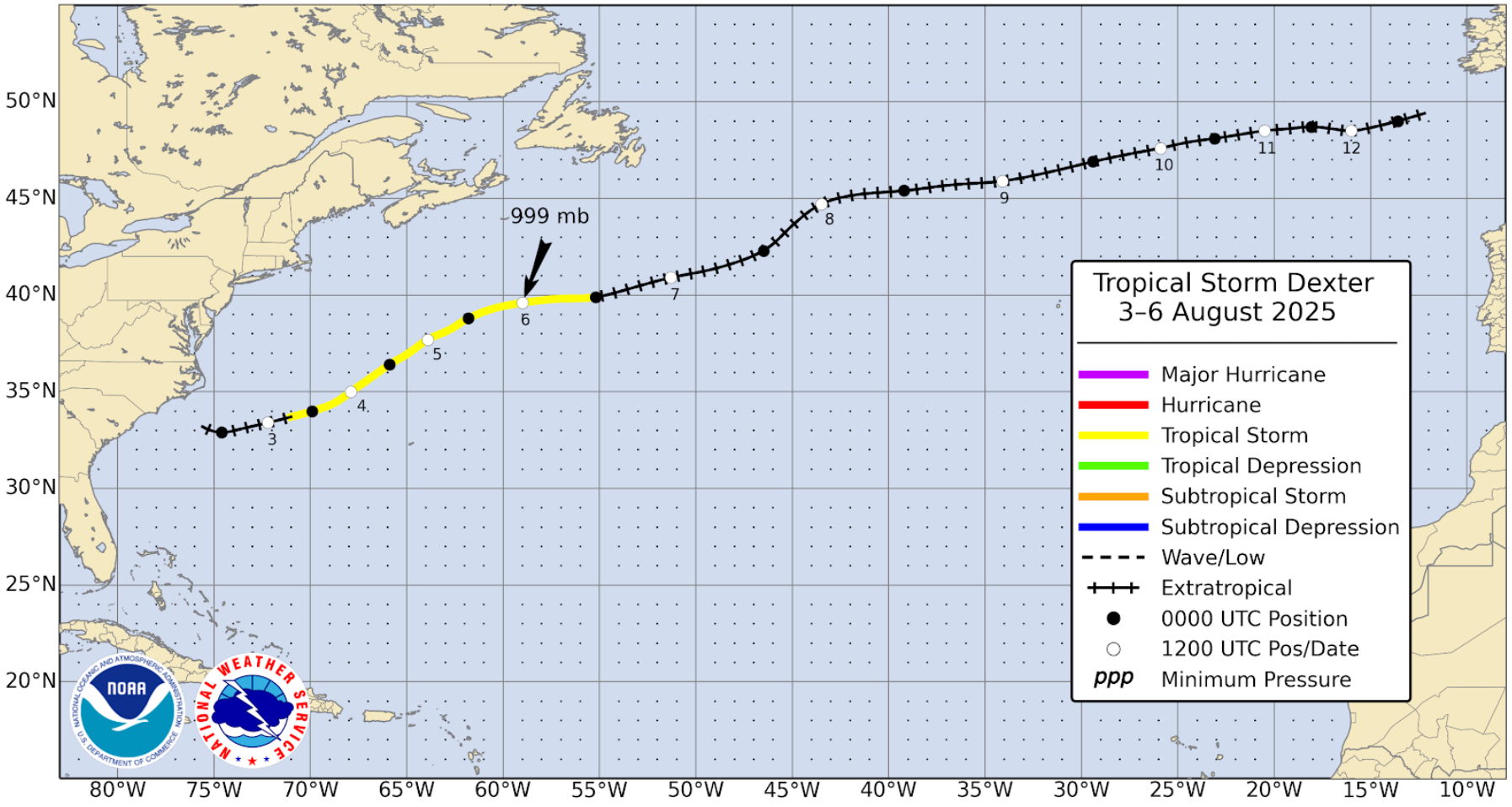


Figure 1. Best track positions for Tropical Storm Dexter, 3–6 August 2025. The track during the extratropical stage is partially based on analyses from the NOAA Ocean Prediction Center.

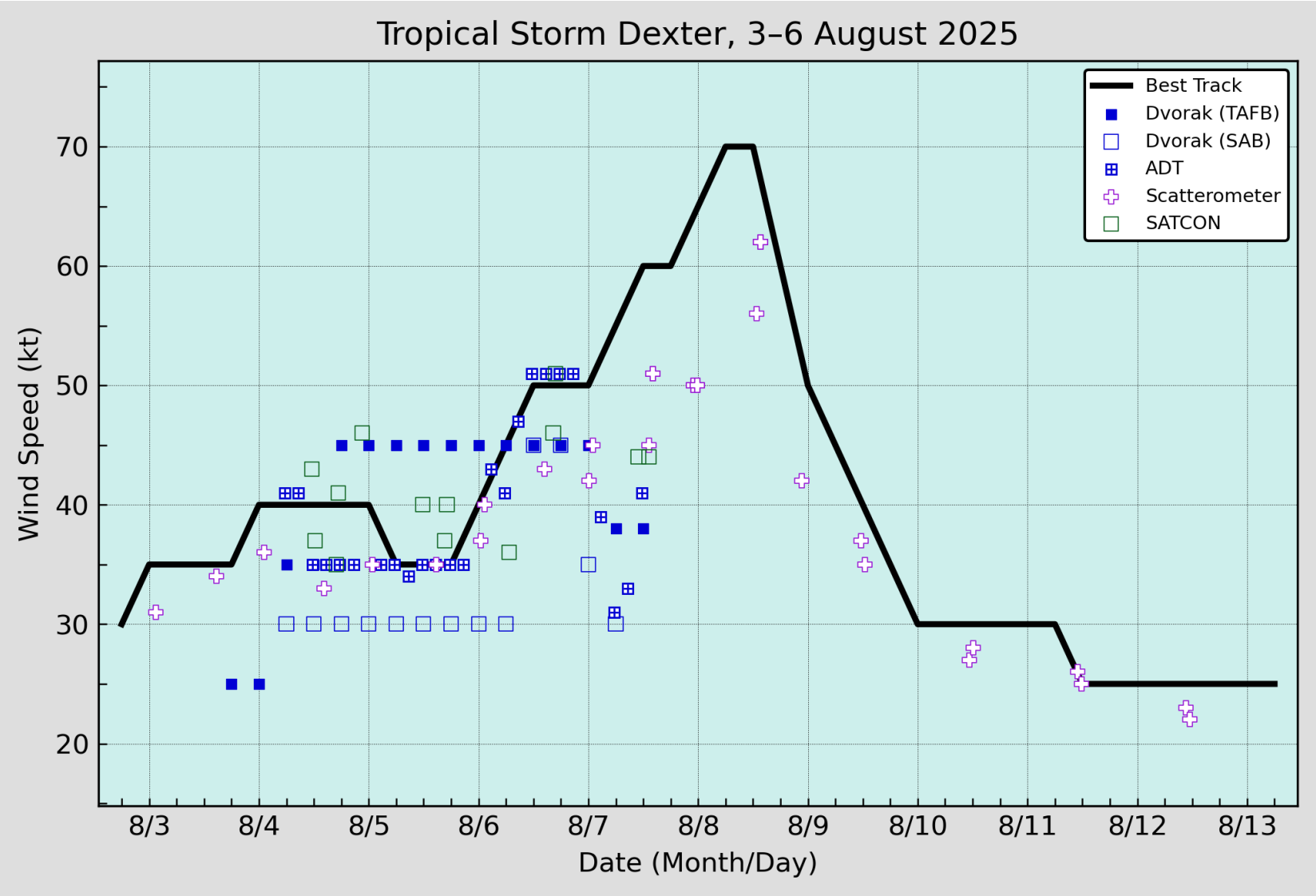


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Tropical Storm Dexter, 3–6 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

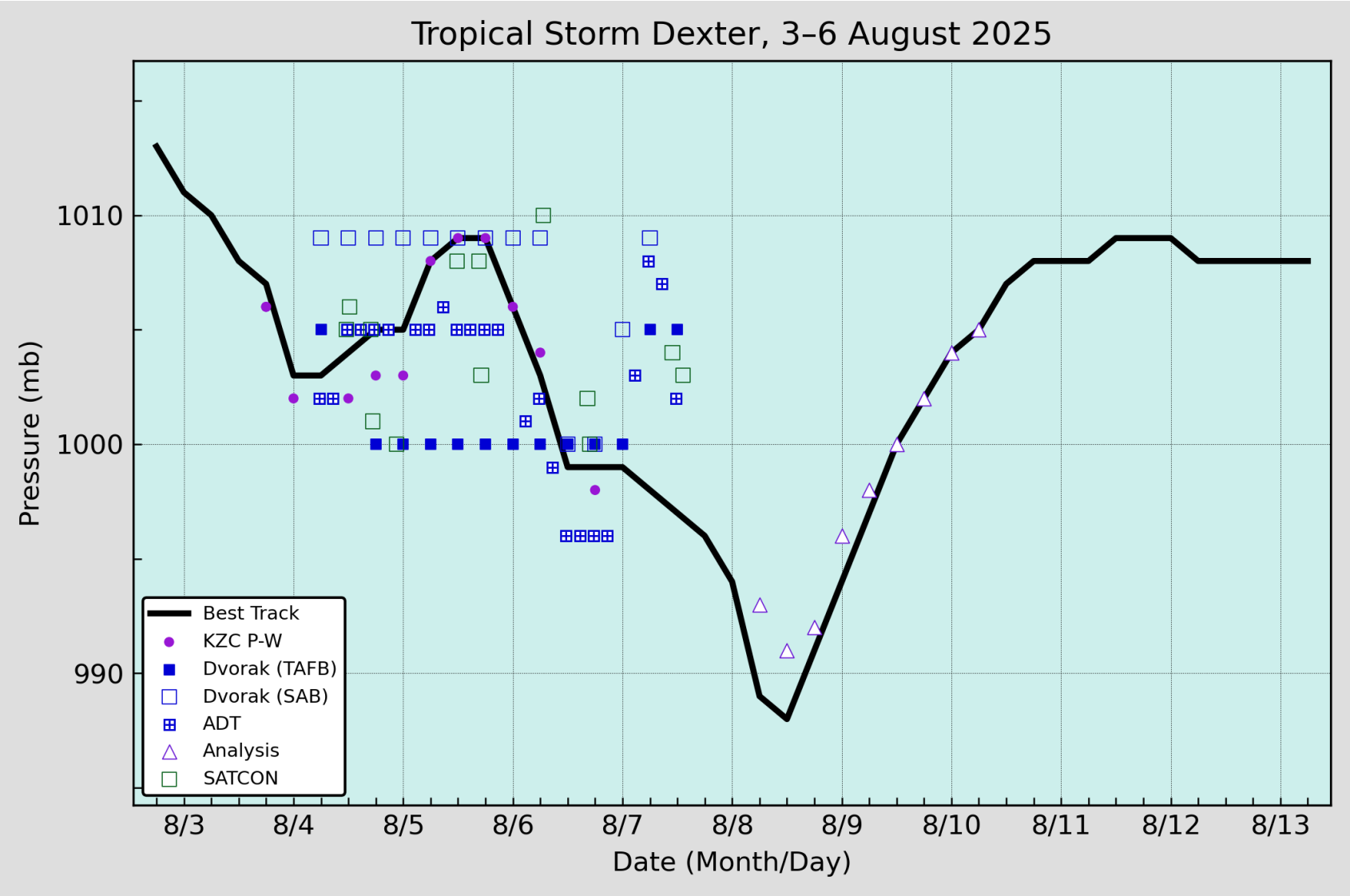


Figure 3. Selected pressure observations and best track minimum central pressure curve Tropical Storm Dexter, 3–6 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

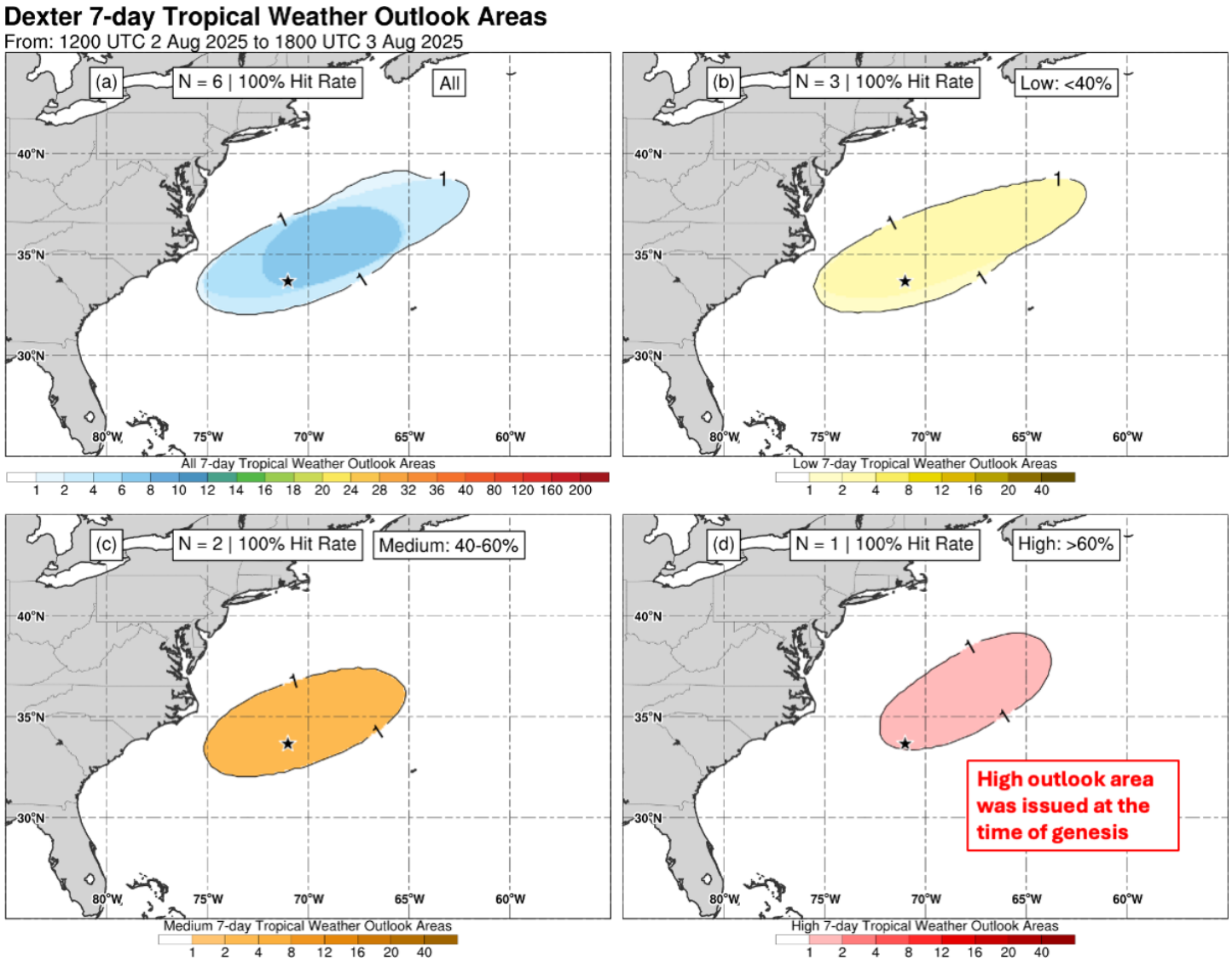


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Tropical Storm Dexter for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.